

POKHARA UNIVERSITY

Level: Bachelor

Semester: Spring

Year: 2025

Programme: BCSIT

Full Marks: 100

Course: Internet Technology II (Programming)

Pass Marks: 45

Time: 3 hrs.

Candidates are required to answer in their own words as far as practicable. The figures in the margin indicate full marks.

Section “A”

Very Short Answer Questions

Attempt all the questions. [10×2=20]

1. What is server-side scripting? Give an example.
2. Write a ternary operator expression in PHP to return "Pass" or "Fail" based on a mark.
3. What is the difference between associative and indexed arrays?
4. Define \$_SERVER super global. Mention one use case.
5. Write a PHP code to store a logged-in username in a session.
6. What is the purpose of the mysqli_connect () function? Write an example.
7. List any five benefits of using OOP in PHP.
8. What is the main benefit of using a PHP framework like Laravel in your project?
9. List any two common input validation techniques in PHP.
10. How does the View differ from the Model in MVC architecture?

Section “B”

Descriptive Answer Questions

Attempt **any five** questions. [5×10=50]

11. Explain different types of operators in PHP. Write a PHP script using form inputs to perform basic arithmetic calculations.
12. Explain different types of conditional statements in PHP. Write a PHP program to take an input number and check whether it is positive, negative, or zero using conditional statements.
13. Explain the concept of functions in PHP. What are the benefits of using functions in a PHP? Provide examples of user-defined and built-in functions.
14. Explain form handling in PHP. Write a program to take a number input and display whether it is prime or not. Validate if the input is numeric.
15. Describe session-based authentication in PHP. Write a PHP script to demonstrate login functionality using sessions and redirect unauthorized users.
16. Write a PHP script to perform CRUD operations with MySQL and explain the flow. The database name is “**program**” and table name is “**users**”.

Group “C”

Long Answer Questions

Attempt **any two** questions. [2×15=30]

17. Explain the core principles of Object-Oriented Programming (OOP) in PHP. For each principle, provide a detailed explanation and discuss its benefits. Additionally, explain the purpose of magic methods
18. What is MVC architecture? Describe the role of each component in a PHP framework. Provide a step-by-step guide to creating a blog post module using MVC (Model, Controller, and View)
19. Define file permission transaction management. Write a program to upload a file with validation and handle error messages if the file is not allowed or exceeds the size limit.